

Pranav Sakhuja

Curriculum Vitae

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EDUCATION

University of California, San Diego <i>Bachelor of Science, Mechanical Engineering with a specialization in Controls and Robotics</i> GPA: 3.76/4.00	San Diego, CA Awarded June 2025
Carnegie Mellon University <i>Master of Science, Mechanical Engineering - Research Track (Incoming Student)</i>	Pittsburgh, PA Starting Fall 2026

SKILLS

Programming: Python, MATLAB, Arduino C, Bash, Linux Systems, Robot Operating System (ROS and ROS2), SQL
Software Tools: Simulink, SOLIDWORKS, Autodesk Fusion, AutoCAD
Manufacturing: FDM/SLA 3D Printing, CNC and Manual Machining, Laser Welding (TPU), CAM, DFM, GD&T
Coursework & Technical Knowledge: Control Theory, Path Planning & Estimation, Machine Learning, Project Management

RESEARCH EXPERIENCE

Morimoto Lab, UC San Diego <i>Research Assistant</i>	San Diego, CA May 2023 – Present
- Developing a minimally invasive optical-biopsy endoscope using a soft, tip-growing vine-robot body, integrating a millimeter-scale camera and rotating polarizer setup that incorporates Morpho-Enhanced Polarized Light Microscopy (MorE-PoL) for colonoscopy applications - Optimized manufacturing process to fabricate a 1.5m long, 12mm diameter vine robot endoscope body with 2 retraction channels using a novel laser welding technique for Thermoplastic Polyurethane (TPU) - Integrated a mechanically programmable, origami-inspired vacuum-driven actuator to enhance the steerability of soft tip-growing vine robots - Developed a ROS-based teleoperation system enabling control of a MATLAB-simulated Concentric Tube Robot (CTR) using the da Vinci Research Kit (dVRK) master tool manipulator - Designed and manufactured a new CTR actuation system, including benchmarking CAN/EtherCAT motor controllers to meet system performance requirements	

WORK EXPERIENCE

KPIT Technologies Inc. <i>Mechanical Engineering Intern — Vehicle Engineering & Design Team</i>	Novi, MI June 2024 – September 2024
- Conducted in-depth benchmarking studies of Thermal Management System (TMS) architectures for various Battery Electric Vehicles (BEVs) - Developed advanced plant models using MATLAB, Simulink, and Simscape, enhancing simulation accuracy and enabling comprehensive analysis of BEV-TMS performance - Optimized various control architectures and schemes, resulting in improved thermal efficiency and system responsiveness for BEV applications under harsh weather conditions by integrating TMS models with powertrain performance models	

LEADERSHIP AND ACTIVITIES

Triton NeuroTech at UC San Diego <i>President</i>	San Diego, CA November 2022 – June 2025
- Directed the management, technical advising, and timeline execution for 12+ research-focused Neuroengineering, Brain-Computer Interface (BCI) projects, involving over 200 members - Conducted technical workshops on data acquisition, signal processing, hardware architecture design, enabling the development of biosensing devices using EEG/EMG signals - Led the organizing team for the 2025 California Neurotechnology Conference, hosting 330+ attendees and featuring high-impact speakers, researchers, and leaders from academia and industry - Previously led the organization as Vice President of Projects and Mechatronics Team Lead for the Robotic Prosthetic Arm Project	

TEACHING EXPERIENCE

University of California, San Diego

San Diego, CA

Supplemental Instruction Leader - Physics

September 2022 – December 2022

- Led collaborative active-learning workshops and peer-learning sessions for 20+ students in PHYS 1A (Mechanics), reinforcing kinematics, Newton's laws, energy/momentum, rotation, and fluids.
- Created problem sets, concept checks, and solution walkthroughs that improved exam readiness and confidence on core topics (e.g., free-body diagrams, energy methods).
- Collaborated with course staff to align content with lectures; trained in leadership, communication, and inclusive facilitation to drive active participation.

POSTERS AND PRESENTATIONS

EMG-driven Neural Prosthetic Robotic Arm

Poster: California Neurotechnology Conference 2023

EEG-SSVEP-driven Remote Controlled Car

Poster: California Neurotechnology Conference 2025

PROJECTS

Soft Robotic Optical Biopsy Endoscope

Morimoto Lab, UC San Diego

- Optimized manufacturing process to fabricate a 1.5m long, 12mm diameter vine robot endoscope body with 2 retraction channels using a novel laser welding technique for Thermoplastic Polyurethane (TPU)
- Designing and implementing a camera module, incorporating a dexterous rotating polarizer mechanism for minimally invasive cancerous tissue detection

Autonomous Autogyro UAV

MAE Senior Capstone, UC San Diego

- Led design and implementation of wiring harness, flight controls, autopilot integration, and Ground Control System for an autonomous autogyro UAV capable of real-time surveillance, navigation, control, and computer vision
- Optimized payload integration and flight dynamics to support onboard camera, telemetry, and data acquisition modules

Autonomous Robotic Car

MAE 148, UC San Diego

- Integrated NVIDIA Jetson, camera, LiDAR, and image-perception neural networks for vehicle navigation and obstacle avoidance using ROS 2
- Implemented Visual SLAM for real-time perception and object search in unknown indoor environments

Neural Prosthetic Robotic Arm

Presented at California Neurotechnology Conference 2023

- Designed and developed an EMG-controlled prosthetic arm using SOLIDWORKS, followed by 3D printing and assembly
- Led mechatronic system design by programming Arduinos, signal processing in Python, soldering circuitry, implementing serial communication for motor control with NVIDIA Jetson-hosted ML pipeline

VOLUNTEER AND OUTREACH EXPERIENCE

CREATE & Girls Inc. Eureka! STEM Program

San Diego, CA

Workshop Volunteer

Summer 2025

- Volunteered with Morimoto Lab to host a two-week series of interactive robotics, soft-robotics, and haptics workshops for 30 middle-school girls from underrepresented groups as part of the Girls Inc. Eureka! STEM program
- Delivered hands-on demonstrations of vine robots and haptic devices; introduced CAD/3D printing, Arduino + programming basics, and soft-robotics fundamentals
- Coordinated with faculty, graduate students, and undergraduate volunteers to prepare and run demos, and led Kahoot review activities for both returning and first-time cohorts

HONORS AND AWARDS

- *Runner-up for best BCI project award, California Neurotechnology Conference 2025*
- *Triton Research & Experiential Learning Scholarship (TRELS), April 2024 (Declined offer)*

ONLINE COURSES AND CERTIFICATIONS

Introduction to Engineering Mechanics

Offered by Georgia Institute of Technology via Coursera

Introduction to Mechanical Engineering Design and Manufacturing with Fusion 360

Offered by Autodesk via Coursera